

## Circles and circle theorems - Set 2

TMUA practice questions by topic. Specification MM3.2, MM3.3, M5.8, M5.9. 5 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the complete set of values of  $k$  for which  $x^2 + y^2 + 2kx - 4y + 3k + 4 = 0$  represents a circle.
- A**  $k > 3$   
**B**  $0 < k < 3$   
**C**  $k < -3$  or  $k > 0$   
**D**  $k < 0$  or  $k > 3$   
**E**  $-3 < k < 0$   
**F** all real  $k$
- 2 The line  $y = x + c$  is a tangent to the circle  $(x - 1)^2 + (y - 2)^2 = 8$ . Find the possible values of  $c$ .
- A**  $c = -3$  or  $c = 5$   
**B**  $c = 1 \pm 2\sqrt{2}$   
**C**  $c = 1$  or  $c = 5$   
**D**  $c = -1$  or  $c = 3$   
**E**  $c = \pm 4$   
**F**  $c = -5$  or  $c = 3$
- 3 Find the length of the tangent from the point  $(7, 1)$  to the circle  $x^2 + y^2 - 2x - 4y - 4 = 0$ .
- A** 3  
**B**  $\sqrt{28}$   
**C** 6  
**D**  $\sqrt{26}$   
**E**  $\sqrt{37}$   
**F** 5

- 4 Two circles have equations  $x^2 + y^2 = 25$  and  $(x - 8)^2 + y^2 = 9$ . Which of the following statements is/are true?

- I The circles touch externally.
- II The circles intersect at exactly two points.
- III The point  $(5, 0)$  lies on both circles.

- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

- 5 The point  $P(x, y)$  moves so that its distance from  $(3, 0)$  is twice its distance from  $(0, 0)$ . The locus of  $P$  is a circle. Find its radius.

- A  $\sqrt{2}$
- B 3
- C 2
- D 4
- E  $\sqrt{3}$
- F 1